# Application: Statistics in the Real World

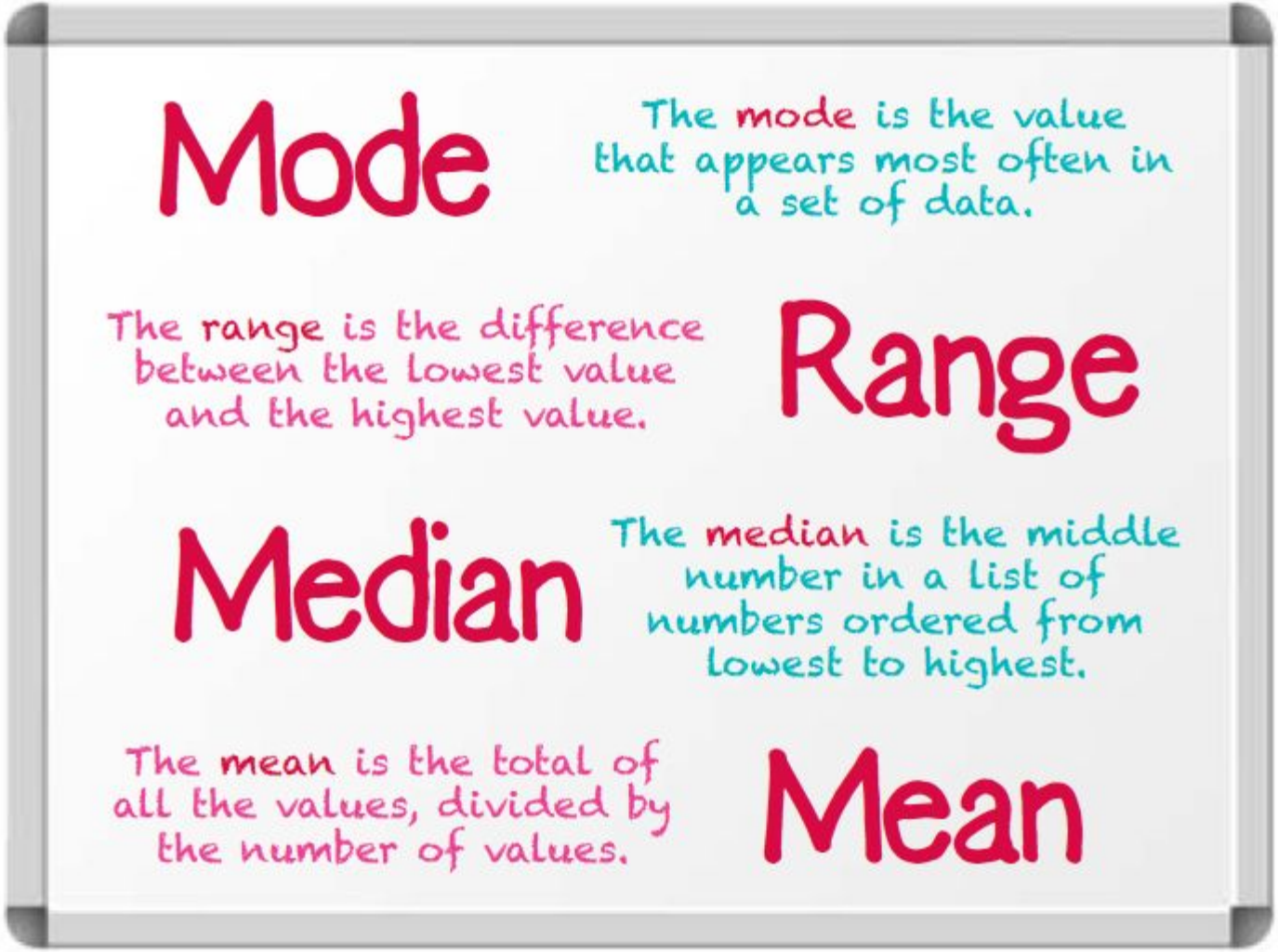


# Review

- Data Preparation
- Data Visualization
- Statistics Terminology and Key Ideas
- Hypothesis Testing
- Linear Regression

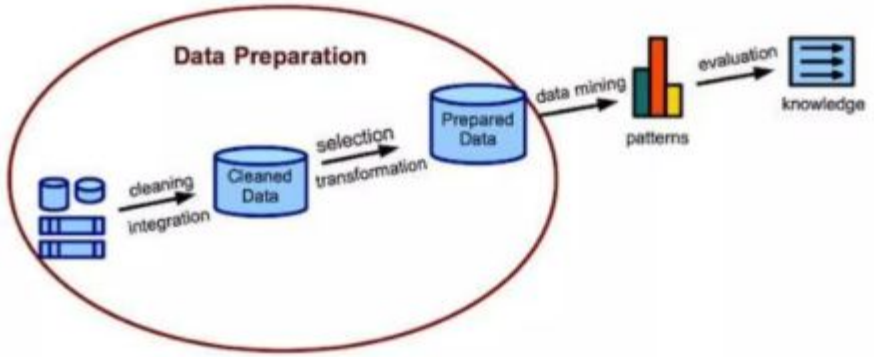
# Statistical Terminology and Ideas

- Mean
- Median
- Mode
- Quantiles
- Range
- Frequency



# Data Preparation

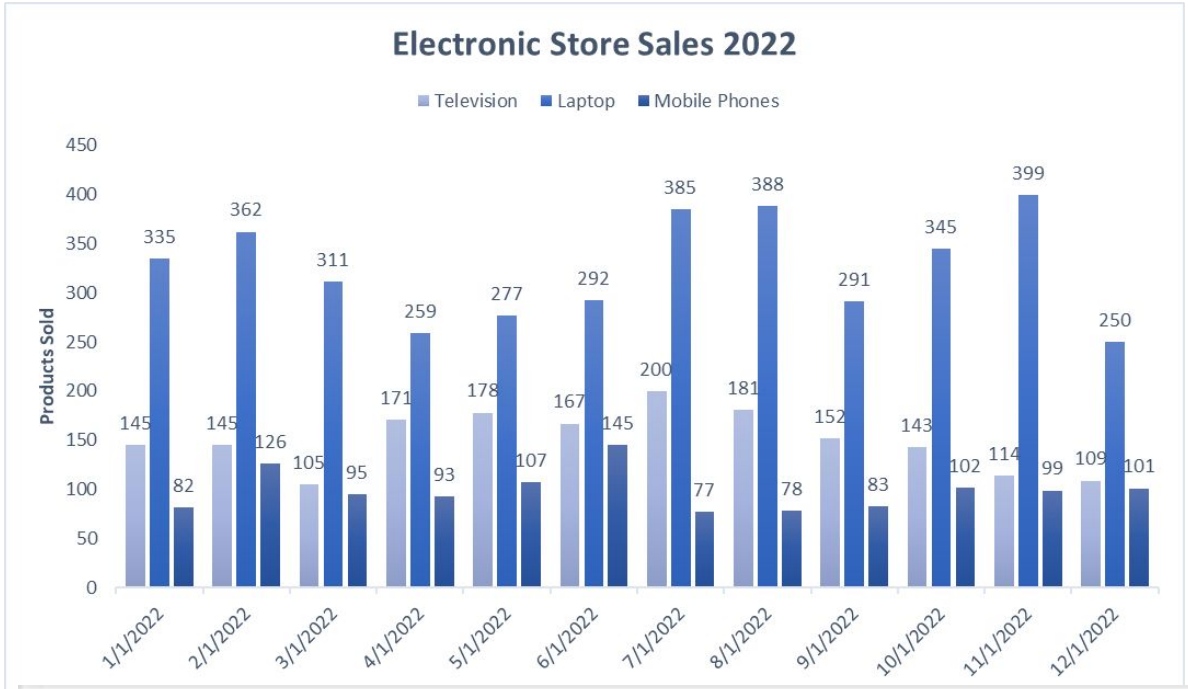
- Data Cleaning: Removing NAs/Unknown values
- Removing unrealistic values
- Selecting certain columns to use
- Use of Binary variables

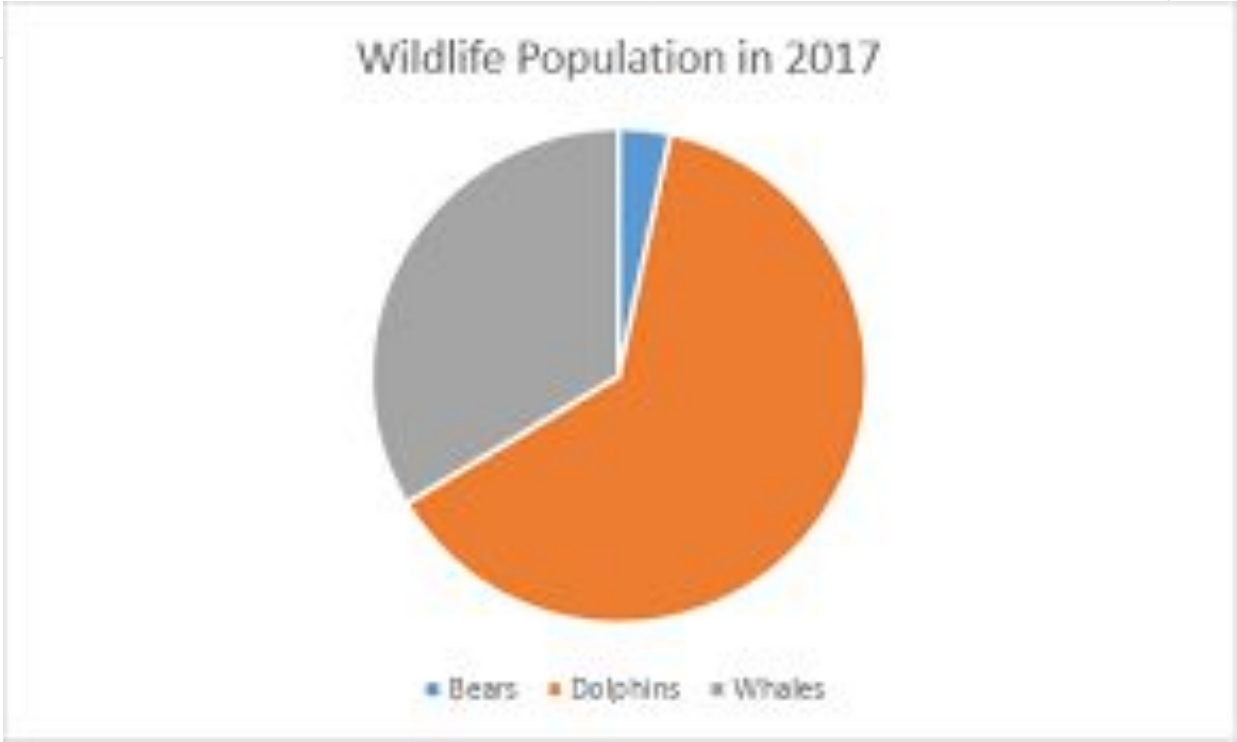



- There are multiple values that are commonly used to represent the virus. A virus like COVID-19 could be represented by 'SAR-Cov2', 'Corona', 'Covid' or 'Covid-19' to name a few.
- A data preparation tool could be used in this scenario to identify an incorrect number of unique values (in the case of virus, a unique count greater than suitable number in case of covid would raise a flag, as there are only few names aligned with virus). These values would then need to be standardized to use only an abbreviation or only full spelling in every row.

# Data Visualization

- Bar Graphs
- Line Graphs
- Pie Charts
- Axes Labels and Main Titles
- Legends



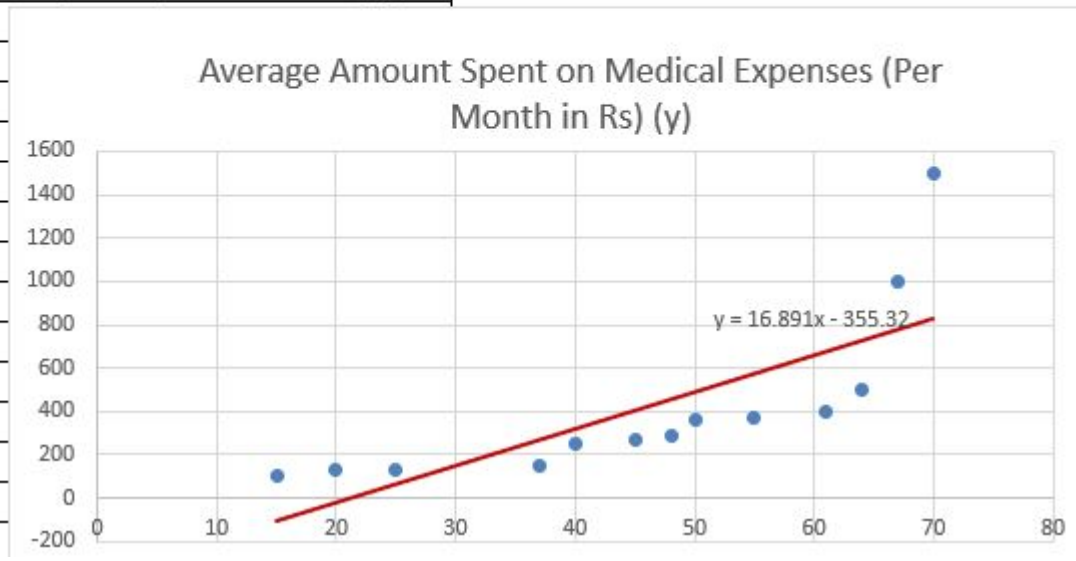



# Linear Regression

- R-Sqaured Value
- Goodness of Fit
- Slope and Y-Intercept
- Positive v Negative Relationship
- Multiple Linear Regression: Multiple predictors for your response variable

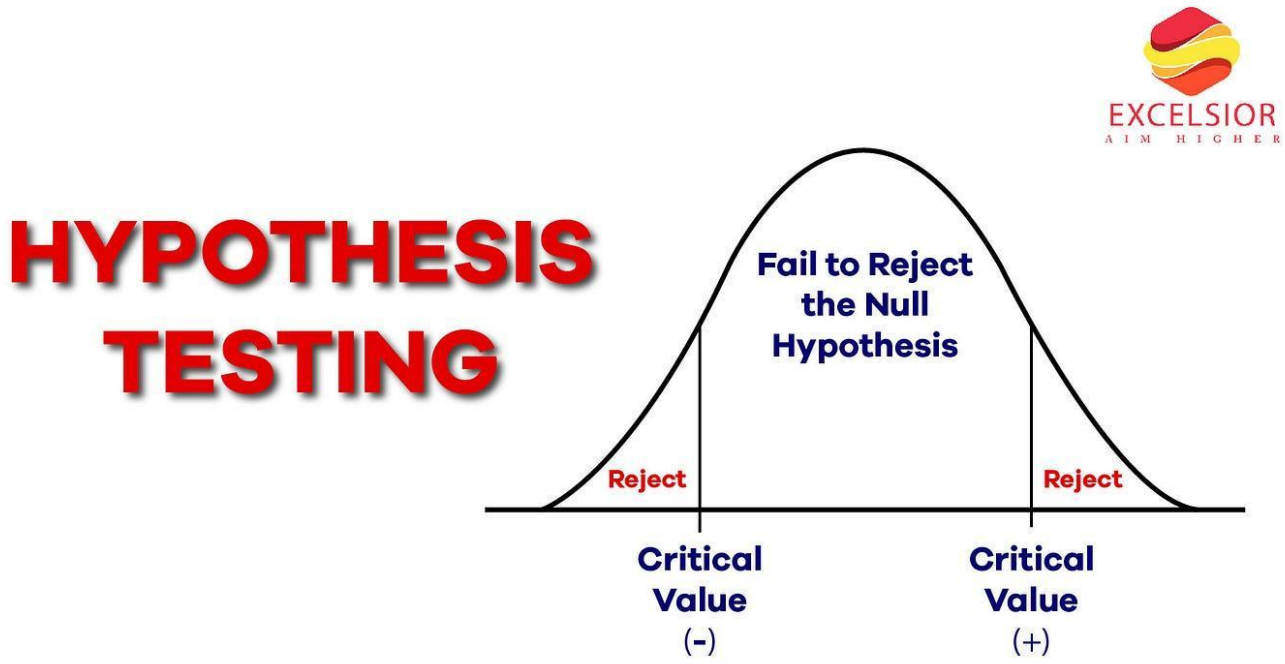
## Linear Regression in Excel

| | A | B |
|---|---|---|
| 1 | Age (x) | Average Amount Spent on Medical Expenses (Per Month in Rs) (y) |
| 2 | 15 | |
| 3 | 20 | |
| 4 | 25 | |
| 5 | 37 | |
| 6 | 40 | |
| 7 | 45 | |
| 8 | 48 | |
| 9 | 50 | |
| 10 | 55 | |
| 11 | 61 | |
| 12 | 64 | |
| 13 | 67 | |
| 14 | 70 | |
| 15 | | |



# Hypothesis Testing

- P-Value
- Significance Level
- One Sample Z-Test
- Z-Statistic
- Null v Alternative Hypothesis



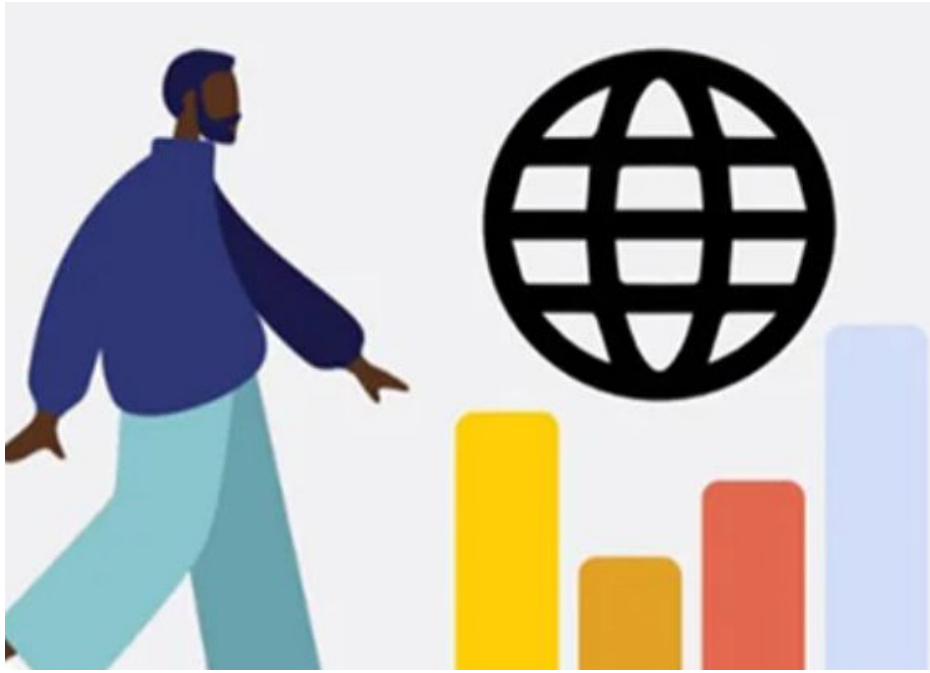
**Can you think of some examples of when you have seen statistics/data analytics in your daily life?**

# Here are a few!

**Advertisements:** When you see an advertisement claiming that "4 out of 5 dentists recommend a particular toothpaste," it is using statistics to persuade you. The advertisers conducted a survey or study and found that a majority of dentists (80%) recommend the toothpaste. This information is presented to make the product appear more reliable and effective.

**News Reports:** News outlets often use statistics to provide context and support their stories. For example, when reporting on unemployment rates, they might say that "unemployment has dropped by 2% over the last six months." This statistic helps readers understand the scale and impact of the change in employment.

. **Digital Content:** On social media platforms, statistics are often used to emphasize the importance of an issue or to generate interest. For instance, a post might say, "Did you know that on average, people spend 2 hours and 22 minutes on social media every day?" By presenting this information as a statistic, the post aims to grab your attention and potentially encourage a discussion about social media usage.



Many fields use statistics for different purposes, such as to help to keep us safe, improve our health, and advance our knowledge.

# Example!

You were just selected as Mayor of a new town! Your town's population is 10,000 people. What ways will you apply statistics into developing your town? How can you apply data analytics to ensure your town's schools are large enough? How will you use data analytics to ensure your town runs successfully?

# Week 9

Error analysis

# Outline

1. What is error and where does it come from?
2. What are the consequences of errors in business analytics?
3. How do we quantify error?
4. What can we do with error analysis?

# Consider the following…



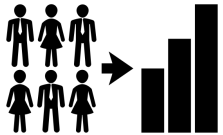
You work for a business selling plants and want a better understanding of your customers:

- What kinds of people are interested in buying plants?
- Can you recommend products to customers that they are most likely to purchase?



You've collected some data by surveying several hundred people and recording:

- Demographic information (age, gender, income)
- Habits and preferences (time they wake up, how much they like animals, how often they buy groceries)
- Which of your products they are likely to buy (none, large houseplant, small cactus, both)



You've "cleaned up" and analyzed the data:

- Calculated summary statistics (average age of cactus-buyer, range of shopping frequencies)
- Used regression to classify if a person will buy a plant as a function of animal opinion and income
- Performed hypothesis tests to determine if there is a significant difference in wake-up time between people who will or will not but a plant



Now what? The next step is to understand how well your data and conclusions agree with the "truth."

*1. What is error and where does it come from?*

# Where and when can error come from?



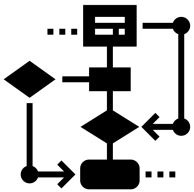
1. Asking the question
   - No relationship between variables you are modeling
   - Poor conditioning: very small changes in x lead to large changes in y
   - E.g. None of the data from your surveys affects plant purchasing



2. Collecting data (measurement error)
   - Imperfect measurement instruments, random “noise”
   - Inaccurate reporting or lack of precision of data
   - E.g. Income is recorded in bins rather than exact amounts



3. Analyzing and modeling data (statistical error)
   - Type I and II errors (false positive and false negative predictions)
   - Uncertainty and variance of random variables
   - E.g. Your model classifies people who rate animals 3/10 as never buying plants, but in reality 25% of them do



4. Drawing conclusions (logical or reasoning error)
   - Assuming causation from correlation, ignoring potential “confounding” variables
   - Over-generalization of results
   - E.g. Your survey was only conducted on residents of one city but conclusions are applied to all people

*2. What are the consequences of errors in business analytics?*

# Bias can lead to decisions that unfairly affect certain groups

Bias = *systematic* tendency of data collection & analysis leading to inaccurate results



- Sampling/response bias – respondents don't represent intended population
  E.g. You want to survey all customers but only your top customers answer the survey
- Social-desirability bias – respondents answering questions dishonestly
  E.g. People report that they are younger than their true age

When unaccounted for, bias can lead to not just inaccurate decisions but also unfair treatment of groups of people and discrimination

- For example: In 2014, Amazon built an algorithm to automate hiring decisions, using past existing employee resumes to train the model. However, the model "learned" to discriminate against women for technical jobs, since most of the current employees were men and the model prefer language men use and reject terms like "women's rugby team"

Do you know any other real-world examples or possible consequences of bias?

# Other kinds of error can also have negative consequences, depending on the context!

Sometimes, false positives are more harmful than false negatives:

- When identifying customers that can profit your business, investing in a non-profitable customer is more costly than missing a profitable one
- In the judicial system, it is usually worse to convict an innocent person than to not catch a guilty one

Other times, false negatives are more harmful than false positives

- When testing for HIV infection, sending a healthy patient for further testing is less dangerous than letting a patient with the virus to go untreated
- In security screening, the risk of allowing a potential threat to go undetected outweighs the inconvenience of delaying a low-risk individual

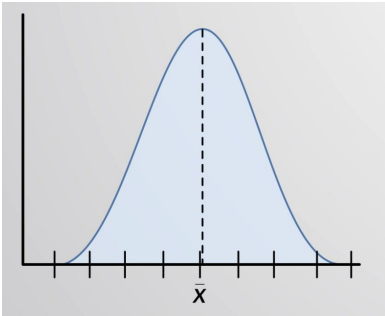
Make sure to think through the consequences of different kinds of error as you make decisions that affect which ones are more likely in your analysis!

*3. How do we quantify error?*

# Quantifying error of a measurement can help convey how confident you are in its accuracy.

*Standard deviation* is the variation of a sample or true population around the mean

Not an error, but rather a descriptive statistic: “How spread out is the data?”




$$\text{standard deviation } \sigma = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

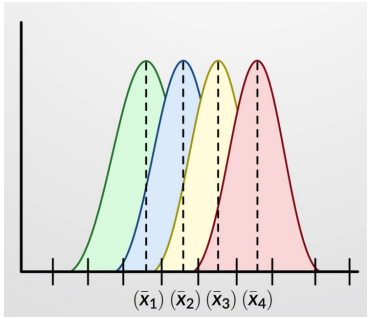
**where:**

$\bar{x}$ = the sample’s mean

$n$ = the sample size

*Standard error of the mean* captures variance of sample means *across* samples

“How accurate is the mean of this sample of the true population mean?”




$$\text{standard error } (\sigma_{\bar{x}}) = \frac{\sigma}{\sqrt{n}}$$

```
= STDEV(A1:A10)/SQRT(COUNT(A1:A10))
```

# Confidence in a model can be quantified with in many ways

## Regression

Report error using metrics like:

$R^2$: “How much variation in the predicted value can the model explain?”

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

| | | Predicted Class: Spam | Predicted Class: Non-Spam |
|---|---|---|---|
| Actual Class | Spam | TP=45 | FN=20 |
| | Non-Spam | FP=5 | TN=30 |

## Classifier

Calculate performance of the model using a *confusion matrix* and deciding what metrics are important for your analysis

| | | Predicted Class: Positive | Predicted Class: Negative | |
|---|---|---|---|---|
| Actual Class | Positive | True Positive (TP) | False Negative (FN) Type II Error | **Sensitivity** $\frac{TP}{(TP+FN)}$ |
| | Negative | False Positive (FP) Type I Error | True Negative (TN) | **Specificity** $\frac{TN}{(TN+FP)}$ |
| | | **Precision** $\frac{TP}{(TP+FP)}$ | **Negative Predictive Value** $\frac{TN}{(TN+FN)}$ | **Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$ |

*4. What can we do with error analysis?*

# What to do with error analysis?

- Identify the scope of your analysis
  - What groups of people does it apply to?
  - How confident are you in your conclusions?

- Iterate to improve and evaluate your analysis!
  - Adjust analysis for systematic errors by collecting data from underrepresented groups
  - Reduce random error by collecting data from more samples

# Week 10

Communicating your findings

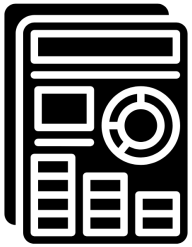
# Outline

1. *Types and formats of communication*
2. *Elements of data visualization*
3. *Tips and tricks*

# *1. Types and formats of communication*

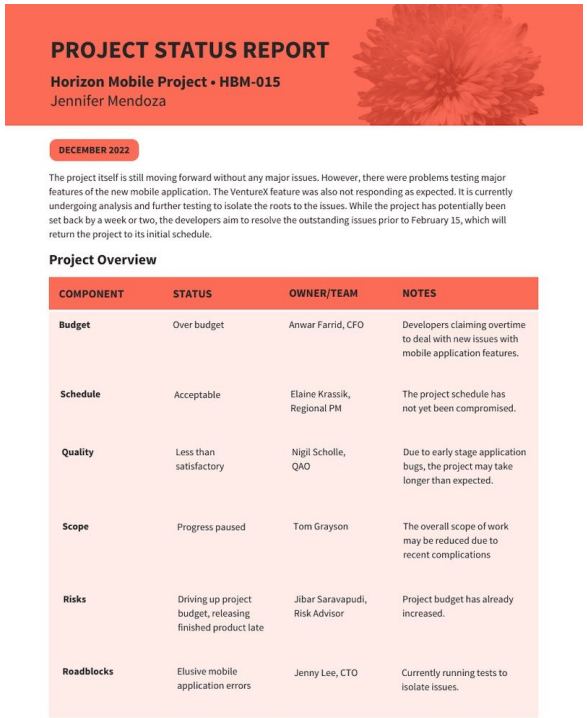
# How might you be asked to share your analysis?





| | **Presentation** | **Report** |
|---|---|---|
| **Purpose** | Engage audience and convey specific message | Provide detailed account of information and findings |
| **Format** | Verbal with visual aids | Document (written/visual) |
| **Structure** | Can vary depending on message, employs storytelling techniques | Usually follow standardized structure |
| **Detail** | Include minimal text and data to make specific point | Include comprehensive account of data, methods, and conclusions |

# Example structure of a business report



**PROJECT STATUS REPORT**

**Horizon Mobile Project • HBM-015**

Jennifer Mendoza

**DECEMBER 2022**

The project itself is still moving forward without any major issues. However, there were problems testing major features of the new mobile application. The VentureX feature was also not responding as expected. It is currently undergoing analysis and further testing to isolate the roots to the issues. While the project has potentially been set back by a week or two, the developers aim to resolve the outstanding issues prior to February 15, which will return the project to its initial schedule.

**Project Overview**

| COMPONENT | STATUS | OWNER/TEAM | NOTES |
|---|---|---|---|
| **Budget** | Over budget | Anwar Farrid, CFO | Developers claiming overtime to deal with new issues with mobile application features. |
| **Schedule** | Acceptable | Elaine Krassik, Regional PM | The project schedule has not yet been compromised. |
| **Quality** | Less than satisfactory | Nigil Scholle, QAO | Due to early stage application bugs, the project may take longer than expected. |
| **Scope** | Progress paused | Tom Grayson | The overall scope of work may be reduced due to recent complications |
| **Risks** | Driving up project budget, releasing finished product late | Jibar Saravapudi, Risk Advisor | Project budget has already increased. |
| **Roadblocks** | Elusive mobile application errors | Jenny Lee, CTO | Currently running tests to isolate issues. |

1. Executive summary: concise take-home message of whole report
2. Introduction: context and motivation
3. Main body: results with supporting evidence
    a. Can be composed of sections for each conclusion
    b. Include data visualizations or figures to support claims
4. Conclusion: recommendations, next steps
5. Appendices: additional data, detailed methods

# Example of a business presentation



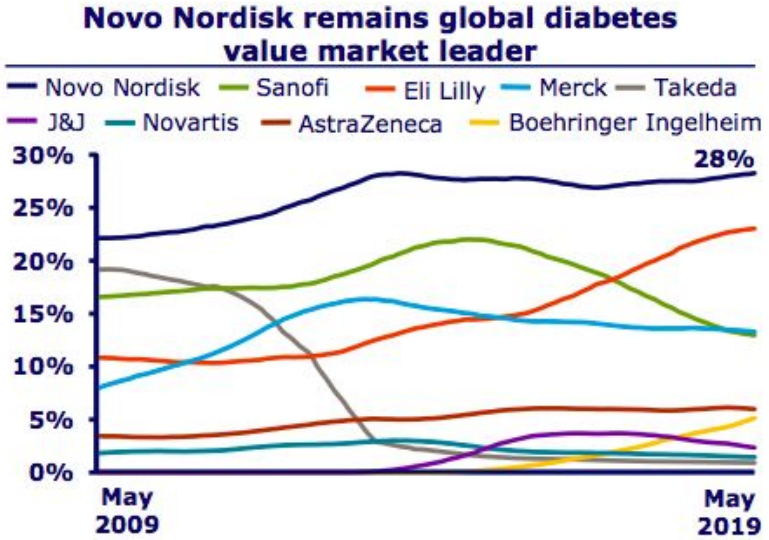

1. Engage audience by setting up the challenge or sharing interesting finding
2. Agenda or outline
3. Main content slides
    a. Use minimal text and strong visuals
4. Key takeaways or recommendations
5. Conclusion referring back to introduction
6. Appendix slides to refer to in case of questions

# *2. Elements of data visualization*

# In presentations and reports, data visualization is key

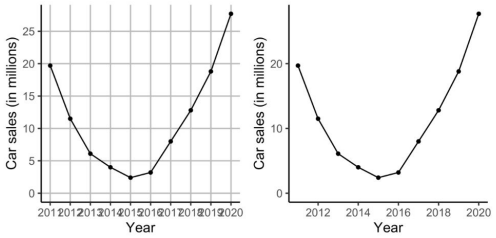
The charts and graphs you use to *analyze* your data might not be the best ones to *communicate* your results!

Data visualization in presentations should directly support your claims:

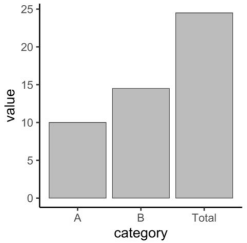


# Ensure graphs are clear, readable, and not misleading:

## Data-to-ink ratio

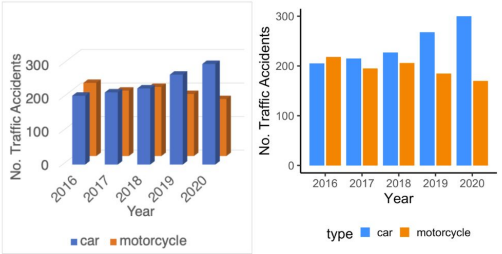



## Data density

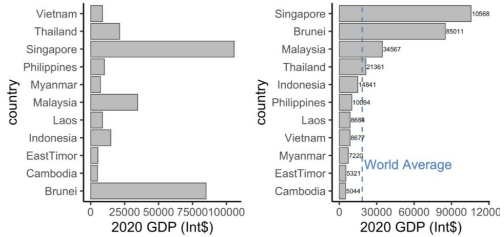



| category | value |
|---|---|
| A | 10.0 |
| B | 14.5 |
| **total** | **24.5** |

## Avoid optical illusions




## Annotate and emphasize

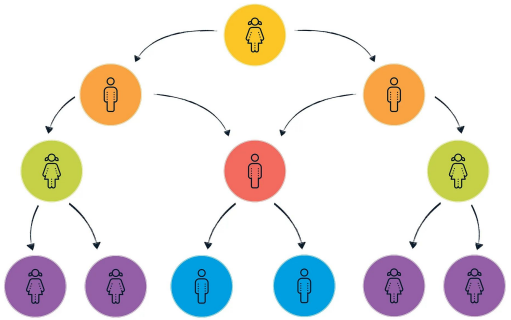



When presenting a graph, tell the audience:

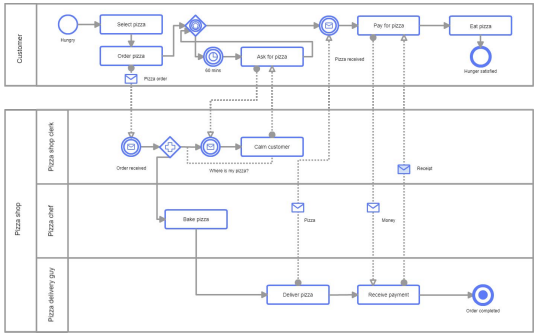
- What does each axis represent?
- What does each line, point, or color represent?
- What is the takeaway from the data?

# Charts are a powerful way to convey more than just numbers

## Organization structure



## Process or workflow



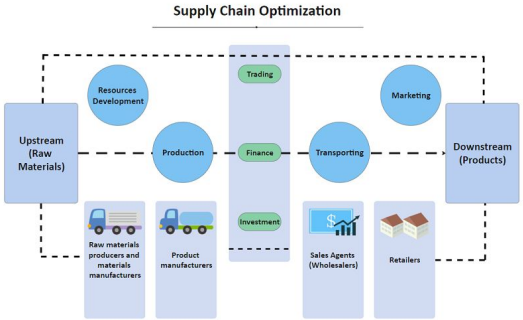

## Venn diagrams




## Gantt chart (timeline)

| Task Name | Q1 2019 | | | Q2 2019 | | Q3 2019 |
|---|---|---|---|---|---|---|
| | Jan 19 | Feb 19 | Mar 19 | Apr 19 | Jun 19 | Jul 19 |
| Planning | | | | | | |
| Research | | | | | | |
| Design | | | | | | |
| Implementation | | | | | | |
| Follow up | | | | | | |

## Supply chain



# *3. Tips and tricks*

# Tips and tricks for communicating your analysis

- The order you did things usually not the best order to tell things!
- Tailor your approach to your audience
- Provide context for numbers: how do they compare to an average or known reference?
- Avoid using acronyms in text or speech
- Be prepared for questions at any point during a presentation

**Tools for slides:** Microsoft Powerpoint, Keynote, Google Slides

**Tools for documents:** Microsoft Word, Google Docs, Adobe InDesign, Canva